

Iain McGilchrist ~ Active Inference Insights 023 ~ Hemispheric Lateralisation, Relationality, Being

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hello everyone and welcome back to a very special episode of active inference insights I your host Darius parvey Wayne and today I'm absolutely thrilled to be speaking to Dr Ian mcgilchrist Dr mcgilchrist is a psychiatrist neuroscientist philosopher and literary scholar best known for his two books or rather Toms the matter in his ersary and the matter with the master in his ersary rather and the matter with things in these he explores radically different worldviews presented by the two hemispheres of the brain and the many implications both at the personal and the societal level of such cereal lateralization in mcil Christ welcome to the show and again thank you so very much for joining me thanks Darius it's a great pleasure so I think um as I said sort of off stream plenty of people in the audience will be familiar with your work and and I'm sure have read at least one of the two books but it might be worth just starting with a very general overview of the kind of lateralization hypothesis um which we can use as a sort of launch pad to get into deeper issues yes well um most people who don't know an awful lot about it um know that this has all been rubbished long ago and that differences between brain hemis pop psychology and and all that and and that that's a position that I understand because after some phenomenal experiments that were made possible by the so-called split brain operation carried out in Caltech in the 60s and 70s to enable people with intractable epilepsy to lead a Life by dividing the two hemispheres it was possible for psychologists to interview and engage with one hemisphere of a person at a time and get quite different results this led to a lot of speculation most of which was wrong but when we start speculating in science we usually get it wrong it's not unusual this is the whole process of science you you you refine it you change it you replace it so um there are three things that make it a curious issue and one worth addressing first of all I suppose before I even say that um any neurologist knows that it's not just the site but the side of the Legion that makes the difference so the site is important but the side is crucial but when you think about it in the abstract the brain an organ which has power from making connections and whose power exists

in connections is divided why it's also asymmetrical in just about everything that can be measured both functionally and physiologically why and and thirdly there is a band of fibers at the base of the brain called the Corpus callosum which connects the two hemispheres only about 2% of neurons actually cross it and a lot of its traffic is in the service of inhibition it's one hemisphere inhibiting the other as much as exciting it so those questions led me to uh 30 years really of examining this question and uh what I think uh the the explanation of this strange phenomenon of the divided brain is can be put like this it answers a simple evolutionary question how to eat without being eaten so in order to get stuff principally to eat but also perhaps in a bird of TG to build a nest in an ape to pick up something to use as a tool you have to be able to bring a narrow beam sharply focused attention to Barrel on a detail but if that's the only kind of attention you're paying then you're not aware of the other 357 degrees Arc of attention available to you and in that arena there may be a predator there may also your kin your mate and your offspring that you need to look after so the one Hemisphere and in humans this is and it becomes so over Evolution the left hemisphere that deals with this very narrowly targeted attention of something it already knows in order to grab it and get it is conducted out of the left hemisphere it's the left hemisphere that controls the right hand with which most of us do the grabbing and getting and the right hemisphere has being left to the task of understanding the whole of the rest of the picture so I mean as a sound bite I sometimes say the left hemisphere um manipulates what's there it apprehends it in the sense of ADprehend grasping onto something but the right hemisphere understands it it comprehends it in other words holds it together so those are two important distinctions and I'll just say a little about the consequences of this when one hemisphere is working alone and of course in the normal situation that doesn't happen but what we know from uh either accidents of nature or from experimentation is that the left hemisphere sees a world uh corresponding to the kind of attention it pays because attention brings into being the reality we experience how we attend matters and if you pay this peace meal ATTention to small bits what you see is a world made up of small bits a fragmented world a world in which things are cut from Context separate from one another uh known because they're the things the left hemisphere targets what it's familiar with what it wants um they are also uh static they're fixed by this Gorgon gaze the thing I want um and they're abstract they're C ized they're decontextualized and effectively inanimate and I'm not using that expression in a loose way we know that there are differences between the hemispheres in the way in which they see things such that the right hemisphere may see things that we normally think of as inanimate as animate for example the

sun and the left normally think of as animate as inanimate for example um your partner uh so uh this has big consequences the right hemisphere meantime is seeing a world in which not that it's chaotic but nothing is ever finally fixed and certain nothing nothing is ever fully known nothing is ever entirely static it's always on the move it's always massively interconnected with lots of other things hence the context Stripped Away by the left hemisphere makes a huge difference to the meaning of what it is we're seeing um it understands what's implicit the left hemisphere more concent on what is explicit and I would argue that much of what gives value to life is implicit and this is an effectly effect a world which is um living in animous and there are just two further things to say one is that the right hemisphere sees a world as it presences uh to use a hyeran term but what it means is as the as whatever it is that we're attending to is coming into being for us whereas the left hemisphere sees a representation which is literally something that is present again after the fact when of course it is no longer present so one is effectively the left hemisphere's World rather like a map a schema a theory a diagram whereas the right hemisphere's world is more like the world that is mapped in all its richness and complexity there that was my elevator pitch in whatever it was five minutes excellent no that's that's very good uh that's excellent elevator pitch um there's a you know at least within the active inference Community there's a there's a general understanding or a general uh claim at least that neuroanatomy has a rather interesting mapping to the structure of the world so for example the fact that the brain is anatomically hierarchical seems to have some uh mapping with the the temporal T you know the time scales at which the world unfolds such that if you have a predictive hierarchy let's say your higher order um Dynamics are going to be tracking the the slower fluctuations in the environment and this is all this all un plays neuroanatomically is there a way in which Beyond pure asymmetry and as you say the brain is neuroanatomically asymmetrical not only in its division uh into the two uh hemispheres but also the actual structure of the hemispheres and you outline this in in the master in his hemispher is there any other the way in which the actual neuroanatomical structure of the brain resembles these underlying functions by function do you mean the world I I mean in so far as if we have these dichotomized roles for the hemispheres yes um so implicit versus explicit the Galt versus uh the part if you will yeah is it is there anything that you know a neuron atomist could look in look inside the hemispheres and say okay there's something uh gen genely physical that is mapping onto those dichotomies uh indeed yes I mean I think the structure of the brain does reflect the if you want to put it this way the reality that with which it's engaged it's hardly surprising because it will have evolved

to understand whatever it is that it's there to engage with and of course the question arises is the structure of the a function of the structure of the world that it's engaging with or is the way we see the world a function of the structure of the brain and I made that point in the presence of ran Williams a former Archbishop of Canterbury and he he replied with a very Migel Christian point that I should have forestalled does it have to be either and I think that's probably right they are interdependent and they reflect one another and the thing that really is I think not only just intellectually interesting but of great philosophical significance is something that I spend the first 400 pages of the matter with things explaining which is that when it comes to the what I call the portals through which we can gain information about the world and and I take these to be and other people might take them to be something else but I think it covers the ground I take it them to be attention in the first case perception which of course is not the same as attention but partly depends on attention um judgments that we form on the basis of what we attend to and what we perceive um emotional and social intelligence and understanding uh cognitive uh intelligence IQ basically and a degree of creativity whereby we adapt what we're seeing according to possibilities in each of these cases the right hemisphere superior to the left and and so and the odd thing is you know well people will say well that can't be right because why would the right hemisphere be always superior to the left what's the point in having it and the answer is the left hemisphere is vastly superior to the right in one thing which I call apprehension which is being able to grasp things and calculate with them and move them about and use them effectively so it has one value which is utility and the right hemisphere has other values um such as you know goodness Beauty and truth so there is a hierarchy both in how good they are at understanding and in what they're able to understand and that hierarchy suggests that the right hemisphere should be if you like in uh overall control not close control but overall command of what the right hemisphere is telling it and it's interesting that from an electrophysiological point of view the left hemisphere communicates with the right hemisphere less frequently and less um uh uh to a lesser extent than the right does with the left there are a number of asymmetries built into the structure and function of the brain and they they point to differences that I think are essential in the world especially for people like philosophers who are trying to um grapple with questions like what can we believe to be the case excellent yes so so the notion of asymetry here is really interesting especially kind of functional asymmetry which which is I guess encoded in the very title the master in his Emissary this kind of N and Fable that that the right is the master and and the left is the Emissary um it maps at least in my mind to a kind of classic you know you mentioned utility

there a classic distinction within psychology and now theoretical neurobiology about exploitation versus exploration and these tradeoffs as I'm sure you know that are fundamental to our sense making and our self-organization and um I've been thinking about this especially in terms of attention because attention seems simultaneously to be a kind of utilitarian process such that we zero in on um things that we want but also it's exploratory and this comes back to kind of the very notion of salience this this distinction between um you know it's Information Gain fundamentally and I've been thinking fun is there an asymmetry there in value and I kind of think there is because I think there's an interesting point that information gain is only useful if it's in the game of providing useful information gain to the organism that then will uh enact exploitative Behavior so um I've been writing this paper it's just been accepted doesn't really matter but I have this example of um a uh a bird versus a kind of Floating Leaf in the periphery of my visual uh field and actually it isn't really worth me paying deep attention to it because to uncover the hidden cause of that is a waste of time and it doesn't yield the same pragmatic effects as it might be the epistemic value however if that object seems to start to engross my attentional field I will you know automatically turn to it and try and uncover its hidden C which implies to me that that epistemic information gain is somewhat enslaved by the utility of getting out of the way of whatever's coming my way all of this to say that is there a potential here is there a potential counterargument that the left hemisphere might itself be the master because everything is you well everything can be summed in some utilitarian sense or or in the sense of utility in terms of pure self-organization and and in terms of predictive processing and active inference providing model evidence for our preferences or just um bringing the work bringing the world that we desire into being and actually all the you know Beauty truth goodness morality and so on are in fact enslaved by that overall goal well naturally you would expect that I don't agree with that at all um it's simply the devil advoca playing you know ramachandran's Devil's advoc yes yes no that's fine of course and we don't live to eat but Eat to Live so the business of getting stuff to help us Thrive is an important part of our life but it's by no means all of our life or the life of any animal so I mean there is a question for the utilitarian which was first pointed out by lesing in the 18th century he has to answer the question utility For What and if the answer is just for further use it hasn't really got us any further forward so utility is a service in in permitting something else to be going on so I mean I would say put it from a philosophical point of view the other way round um but I'd also say that the one that explores which is the right Hemisphere and is therefore able to look around the environment and see what is there you

could say that's in utility because I mean you can reduce everything to a utility if you wish it's just a point of view um but it's not the only point of view um in that point of view it is the right hemisphere that is being just as useful as the left hemisphere which is exploiting what it knows and indeed the left hemisphere won't get very far without having the exploratory attention of the right hemisphere on its own it will it will not survive and and it is interesting that these two types of attention and I'm sure you were putting it forward um in that in realization of that that the right hemisphere is the hemisphere of exploration and the left of going a little bit beyond what it is we know whereas the right sorry the left hemisphere is enslaved to use your term to what it is it does know and therefore is not good at changing set being flexible suddenly realizing I need to look at this differently there's there's a big literature on that on the relative inflexibility of the left hemisphere and when you put that together with its knowing less understanding less um I think you've got a hard case to make that and I know you're not W to making it that the left hemisphere should be viewed as the as the master there one little clinical observation is that when you have frontal damage and the posterior cortex is released from frontal control in the case of the left hemisphere the hand starts grabbing and grasping things even imaginary things but it starts to it will go to a picture and try and take something out of it it tries to lock onto if it's passing a chest of drawers it'll try to catch onto the knob whereas in the case of the right hemisphere when it's disinhibited from frontal control you the hand starts to explore it explores the environment with sort of movement of a tentative kind going out so it is it is I mean there's no question that those are important differences between the two hemispheres quite I mean so I've been really interested in not only exploratory versus exploitative behavior with respect to attention but also so-called endogenous versus exogenous attention uh which is a dichotomy or distinction as you know that goes all the way back to the 80s and and Posner and people like this and and and the you know to cut a long story short the brunt of my paper is I think this dichotomy is massively oversimplified and needs dismantling somewhat although the distinctions are useful but in terms of going back to the neuro anatomy there's a really interesting finding so this is mainly Coretta and Shulman and their big 2002 nature review paper and they find that um endogenous attention which is generally taken to be goal driven and voluntary is localized to the intraparietal cortex and the superior frontal cortex not necessarily with hemispheric specialization but exogenous attention uh which is localized to the temporal parietal cortex and the inferior frontal cortex is mainly lateralized to the right hemisphere and I think this maps exactly what we were saying which is that exogenous attention is constantly being drawn by

interesting and relevant stimuli one thing I wanted to ask you about is one of the classic criteria of um exogenous attention is it's involuntary and I think we we we sense that right if we hear an ambulance siren we're immediately drawn to it and this in principle is an instance of exogenous attention but that doesn't really seem to be in keeping with the right hemisphere's relationality um it's I don't know it involuntary doesn't really seem to be particularly right hemispheric um capacity so so how do you sort of marry that and make that compatible well of course the first thing to say is that involuntary motivations are present in throughout the brain and both hemispheres quite obviously but I don't think the right hemisphere's attention is simply passively drawn if that's what involuntary means towards a stimulus I mean of course if there is a threatening or exciting stimulus then the brain in general is going to be drawn towards it um but the the thing about right hemisphere tension is that it can also calibrate more easily than the left hemisphere whether this attention is as it were um worthwhile is worth the effort of attending to or not so it can inhibit uh one of the things that is important about the right hemisphere is it's capability of inhibiting its own um processes which is a greater capacity than obviously the left hemisphere can do that too but in in general the left hemisphere tends to be more um again to use your useful um term enslave to a response to something which is its automatic response to it and it seems um to jump to conclusions so sometimes people say to me so I guess that um uh oh God what's his name you know Nobel prize winning system one system two I've just forgotten his name um sorry con I mean I I know conman well but sorry his name just disappeared um so caraman system one and system two and and we've corresponded about this very briefly but um we don't think that I mean first of all he's carving the brain from top to bottom more than from side to side which I am but but also because the evidence is that the left hemisphere jumps to conclusions it's quick and dirty it wants the decision fast partly because it is the predator in us and can't really afford to miss an opportunity I mean it may get it it doesn't really matter but but hesitating and you will lose it so it tends to rush in whereas as R chandran says the right hemisphere is the devil's advocate it's losing milliseconds by saying yeah but hang on it could be this it could be that and is much more better at inhibiting what it is it's going to latch on to as well as inhibiting inappropriate emotional responses which is uh interesting because it's well it goes with it being emotionally more sophisticated in the left hemisphere yes I wanted just the very that's all super useful and actually speaking of usefulness back to usefulness um there's a constant tension I guess within the brain sciences and the human Sciences about the distinction between thology and tonomy and thology is a kind of Aristotelian notion that there really is a a final cause and we're moving

towards something in a purposive Manner and in tonomy is the apparent purposiveness of our Behavior behavior and I thought you said something really interesting there which goes back to lesing which is that it's always utility for something and the way that I pose that initial question was the utility is defined by these kinds of preferences which in a predictive processing hierarchy is encoded right up and down a generative model and in principle everything unfolds in the through the lens of of those predictions but I sense the kind of skepticism on your part of that because I mean one part that seems an incredibly uh selfcontained and self-referential system and doesn't really allow in any profound sense for relationality and maybe perhaps the generative model or these preferences should span the agent environment Arena but you know more broadly is there do you do you believe in a in a fundamental Telos for which are attention is being guided or these hemispheres are acting in correspondence Beyond let's say mere survival or mere self-organization well it's a very good and rather rich and complex question I think um I think it depends very much what you mean by um a te loss um in essence as you pointed out um the necessity to survive must be a goal for whatever a creature is doing that that's bound to be part of it um but then you ask about whether there these other values perhaps Beyond and I think that there are I think I think that we we've we've bought into a very hydraulic system in which what we do is push from behind so rather like a mechanism we we're sort of responding to something that pushes us and we we either push back or we go on or whatever but we're not drawn from in front and I think that when you think about the way human lives evolve and why we do any of the things we do we cannot ignore the fact that we are drawn to certain ends certain not that we have a necessary plan of how we're going to do it but those are ends that we we we feel um an Allure to that we we it's esteem we value we wish to um respond adequately to and I think there the right hemisphere is has a broader canvas than than the left in what it's doing and I'm not sure that really the distinction which I'm familiar with between teleology and theonomics is is really getting us very far I I I feel it's a kind of a verbal slight of hand whereby one can reintroduce teleology which has been a big no no for such a long time without appearing to do so and you know but at last we're able to say not that Turtles go ashore and lay their eggs but we can now say Turtles go ashore to lay their eggs they do and it seems it seems it seems particularly perverse and artificial to deny that any creature has any kind of a goal even if you look at an Amber under a microscope in in in a drop of pond water you can see it exploring certain things and gravit towards certain things so I think that there is Direction and that direction is partly governed by our values and not the only value is utility I claim um but there isn't a kind of

what people fear I think completely rightly is the idea of it's all being worked out in advance and this is a purpose that the universe is taking us to that that implies some sort of overall um very large divine kind mechanism a sort of you know Celestial engineer who's worked out the plan and I don't believe that so I think that nonetheless there is shape and meaning to the direction that Evolution has taken I think it's extraordinary that it has managed to invent the eye 14 times according to Robin Conway Morris um because it's a rather amazing thing to have invented even once yes so there's all this wonderful work by Michael Levin I'm not sure whether you're familiar with Mich work about sort of morphogen and the capacity for for these organisms sort of grow frogs legs in or an eye on the Frog's leg and and all this incred like this apparently invisible morphogenetic space in which things seem to tend um and and again I think well there's there's an ongoing debate I mean this is this podcast is called active inference in sites and often times I will diverge from active inference because my fundamental orientation is philosophy and active inference is fundamentally aerior physics but I'm not sure and we can you know I can try and unpick this but there's an ongoing debate in active inference and and the free energy principle which is kl's great um you know architectural achievement which is whether you know things purely look as if they are maximizing model evolence as if they're doing some basian work they're minimizing variational free energy so that's it looks like that's a descriptive instrumental tool for you and I as scientists or whether there is you know things actually have the architectural systems that um might solely be descriptive so this the mark of blanket for example or they are actually doing some form of variational basian inference and I guess this comes very much to our question of Telos andology I mean Ian is it is it there a risk that when I say well actually all I can ever really say is that this is an exceptionally good modeling tool and I can use it to sort of get very close to the phenomena but never really speak of the phenomena how how much of a limitation do you actually see of that and and also do you see that applying also to the hemispheric hypothesis yes I think one has to distinguish between um the of Truth and and whether a certain thing is true to what it is modeling uh and um whether or not it is simply another way of describing it so I would say that hemisphere theory is not just another way of describing it adds something but I think it's generally true that when we model in science as Neil bore said we are not necessarily talking about the reality of what is there what we're talking about is making our observations cohere with with um with what we making them into a coherent picture of what it is we're observing so I in other words it's not really about finally uncovering the reality it's about a way of modeling which may be good or bad and and I think that that

would be the way I generally look at it um I I do do you find do you find that um if if the the fundamental motive is to have an internally coherent system one runs the risk of pure self-enclosure and self-referentiality um absent any ontological grounding I mean if my theory is simply that things look as if they're doing Bayesian inference and I can say everything from the cosmos to an atom is looking like it's doing so well I've got some fundamental axiom that there are things in the universe like an atom and the cosmos but beyond that it seems like I'm just constructing my theory and then building it out from the inside without ever really getting out of the cave um where it is where where is the kind of tether in your in your mind to reality I mean to to put my cards on the table I think it may be phenomena I I think it may be Consciousness because that seems to me to be the kind of ontological Prior everything itself is realized and understood through Consciousness um and there's obviously a massive problem here about how we can get from mathematics to Consciousness and going from the math to the territory or the map to the territory but I'm I'm curious about how you see that mapping and and again is it Consciousness is it mathematics what is the fundamental tether I don't think it's mathematics I mean mathematics is um is of course um enormously useful and beautiful but it it's by Nature um not intended to cover the whole of reality it um I mean this was a mistake that Galileo made when he said that the the cosmos the book of the cosmos is written in numbers well yes it is some of it but also some of it's written in things that frankly are non-numerical and you won't understand them better by doing more math about them I mean there are um people with um particularly people with schizophrenia but sometimes people with right hemisphere damage who want to work out in mathematical terms what it is that um somebody's doing when they are falling in love or making love or whatever and they have the ambition to be able to reduce it all to an abstract system of mathematics I would say that's misguided because our first understanding is not that it's mathematics mathematics is something we can bring to it but like you I completely um would endorse this idea that it's phenomenology we we take in the world in a host of different ways and some of the ways we aim to understand it are through and this is the second part of my three-part book the matter with things epistemology how do we we get closer to truth I think one undeniably is through science other is through the use of reason but uh on their own they can't deal with everything and in fact intuition is a very important part of reasoning well I always say that people who have good intuitions reason better than those who don't and people who reason well have better intuitions than people who don't so they're they're like a pair of twins that should always work together and and in a way they're in the service of in

ination and this is not in any way to sort of um turn one's back on the the the procedural nature of much of science but when you come to look at the way in which great steps forward have been taken in our understanding in science and in mathematics they've often come from a an Insight which is a gestal shift and and from an imaginative people which is not to say that there isn't important stuff to be done at the micro level doing it in a left hemisphere linear procedural fashion yes there there's plenty there I mean one is it's kind of ironic that physicists and mathematicians think that well either the phenomena is maths or the Consciousness could be you know the the manifestations of Consciousness could be mathematically explained given that their predecessors didn't think so whether that's b or B or Einstein um that's one rather ironic thing I this I mean a lot of this really makes me think of I following the master and his ersary which I finished only about a couple of weeks ago and and massively enjoyed so thank you for all the insights first and foremost I've been um I've been digging into draus because I've always had an interest in draus uh and and have read around draus and a bit of draus but never really committed to draus and draus has this really interesting idea that he in his um negation or his argument against John S that activ and it comes back to what you were saying that we always talk about being pushed from behind but not pulled forward that activity has a world to mind causal pattern such that I feel myself drawn through what Merlo Ponte you know would call motor intentionality into situations and I I want to briefly touch on draus and Mar Ponty and haiger because I know you're interested and extremely well read and well versed with these people that haiger as you know says that the the fundamental way of being is not this kind of cartisian Detachment rather it's doign it's being in the world what what does that tell us about Phenom phenomenology what does that tell us about our Consciousness that if indeed it's fundamental way of being is relational and in the world and non rational in the sense of hyper rationality and and unpicking things in the left hemisphere way does that tell us anything profound about our fundamental way of being I think it does yes and and you know I I I would add to the list of philosophers you mentioned Michael Pani um and and also wienstein wienstein made this um very important observation that when you stop moving and stop acting and just stare the world seems to become very strange and and no longer has any kind of meaning but that when you act in it it does again and this is um an important observation and goes with the ideas of people like Meo ponti that our being is embedded in the world and this comes back to my emphasis on the importance of uh both embodiment and context after all the context in which we largely Act is that of our body and our body's engagement with the world and it's through that that we to understand things not by an

Olympian Detachment pretending to be a brain in a vat and just speculating about things in the abstract so to to understand the world one must be engaged in it and I think that recognition is important one of the problems I'm sorry I'm a psychiatrist I come back I find the whole business of schizophrenia arely fascinating if it weren't so as do I as if it weren't such a widespread condition people would be just completely amazed every time they met somebody with schizophrenia because they see the world in a totally different way and um it's not an exact match but the the simplest way of putting it is they have a hypofunctional right hemisphere in which the left hemisphere is in overdrive and so there's a kind of hyper rationalism going on which is actually not going to lead you to truth or sanity as GK chesteron wonderfully said um ad man is not somebody who's lost his reason he's lost everything but his reason and therefore has to reason on the basis of something that he doesn't understand so the schizophrenic sitting there and going somebody's talking to me in this room and there's nobody in this room so where's it coming from looks around and sees that plug socket over there must be coming out of the plug socket and that's wired into my neighbor's house and so so the whole thing is built entirely rationally but on grounds that are not based in experience so um we have to be careful about hyper abstract and Hyper rationalistic formulations not that they can't cast light but that we should be aware that we're not actually substituting or shouldn't be substituting those for the whole of experienced reality you know the thing about the color blue and and Patricia Churchland sort of saying but the color blue just is a wavelength of whatever it is 104 mutus per second so but but actually it isn't that's that's a of almost irrelevant description of it from outside but it blue is nothing if it is not experiential and that of course is not something that can be done through um the the abstract and numerical approach quite I mean I I feel sometimes quite disheartened reading these neuros um well they call themselves neurop philosophers but it's like they never read any David Charmers or or Thomas Nagel but you know they know far more than I do there's um there's so much here and and I I'm also very very interested in self-consciousness um and self-awareness and what that means because it's an extremely misleading term yes but yes it brings me back it brings me back to haiger because as you know and as you've written about haiger is constantly tussling with this subject object distinction so in one of his lectures he he he says that you know the whole of his work is to uproot the cartisian model that has pushed you know held back Philosophy for for 400 years nevertheless it seems to me at least that darine does still have this structure it does still have this in some ways a relationality it's not a dissolving of things into a w like the Buddhist might speak about there is still the there was still at some level the acting

and the acted and yet it's not in this sort of cartesian manner when you think about about um self and other um and we could I'm very very keen to come to schizophrenia because I think this is really a paradigmatic case where it all becomes very interesting but just at the sort of very neurotypical level of being in the world what what what do you what do you senses going on there I mean I presume it's not some cartesian ego homunculus floating you know behind the eyes but what how much can we really disband that dichotomy and how how important is it to still keep something like a subject object distinction yes well the first thing to say can I start from the end of that question and go back to the first part of it um however you yeah um if I can remember uh so I think the the ability to hold distinction without vision is critical to understanding and it's something we seem to have lost so if you make a distinction people run very quickly to the being a complete divide whereas it isn't it's rather like you know the two poles of a magnet they are very distinct neither can live without the other but they are not divided in fact they are completely joined and you know so and if you try to cut one off from the other you just create another the magnet which is like a hologram of the first so the the idea this idea turns out I think to be really it recurs throughout all my philosophical Explorations the need for Union and division to be unified not to be divided so Gerta said the whole business of nature is dividing the United uniting The Divided and the Cy and the deast and so on he is a marvelous piece but effectively he kept coming back to this that we should see something that it seems one to be manifold but see the manifold as being unified at some level and and you know in in the matter of things I have a whole chapter on this idea which is so important in eastern thinking of the relationship between the one and the many so I'm so I don't want to say that before saying anything about division and and Union in the case of self and other and I I think in the I think it's in the master in his Emissary I I give the example you know of the infant that is completely fused with the mother shortly after birth and there is a total inter subjectivity and that the business of the the gradual separation from the mother over the first year is negotiating a distinction which is not of severance so it's actually an it's not a diminishment of something it's an increase in it in much the same way I sometimes use the image of um two two heavenly bodies that are rotating around circulating orbiting around one another and and if for them to have that orbiting relationship they mustn't crash into one another but they mustn't just fly apart and indeed you can say that about a human relationship that it can become over and meshed so that the the self of each party is not preserved or it can become so divided that the two have to part so the the whole thing is about holding this tension between two things without collapsing it into one or the other into mind or body into

self or other but holding the tension and that negotiation which as I say is what happens in the properly functional relationship through the Intercourse between the right ventor medial frontal cortex of the mother and the right ventor medial frontal cortex of the of the infant brings about this ability to be sort of together but not um fused and I think the distinction between the objective and the subjective is a very important one but the idea that the objective as naively supposed that it cuts off every kind of human contribution to the understanding has got to be wrong because we we can't understand things by not being a human as it were that we'll understand them less well from a very peculiar point of view in which everything about us has been cut off that won't lead to a real or true version of what it is but what I think that objectivity means is the ability to inhabit other perspectives on your own so objectivity is really seeing things from as many perspectives as you can and that's very valuable but it doesn't negate the fact that you have your own point of view your subjective position yeah well I think um the work of more modern phenomenologist like like a Dan Sahari for example or a Sean Gallagher would suggest that the very Act of empathy in this very broad sense of being able to take the perspective of another is Downstream on the fundamental for meanus um what people call pre-reflective self-consciousness such that you know my experience is for me and I have a privilege access to my own experience and in doing so I recognize that you do as well and there is this Mutual modeling and shaping um and yet and and yet um and and so inter subjectivity is a component of this kind of subjectivity and and yet it there still seems to me at the fundament a division is the wrong word and I don't want to say division but the fundamental fact that Consciousness experience is always for me and not for you um and this goes back to all the way back to sort of Louis S and P and the work they did with Dan sahav on the fact that in schizophrenia even that famus is still not lost so for insertion is a really useful example here um you know metser talks about the fact that there's there's no sense of agency over that thought or there's no and others have talked about there being no possession over that for but that for is still for the subject whatever that whatever the form that subject takes so given that is there um is there a way in which by the very n by a very nature as conscious beings who have a privileged access only you know only to our own lens to our own perspective we are cast off from others in in some in in the deepest manner possible I don't think that no um so Consciousness is Consciousness is for me best imaged by by water um so this is an image from shelling but I think it works for me that in the water there can be a Whirlpool or many whirlpools and and the first thing to notice is the whirlpools are not in the water they are the water they're not sort of an extraneous object in the water

they're just the form the water takes at the time and those World Pools May be very self involved so that the movement all which returns within this this flow um it's very real it can be photographed it can be measured it can have power to move stones and it will eventually move on and then it becomes part of the rest of the stream now I see Consciousness the individual Consciousness as being somewhat like this that it is connected and in an enormous number of experiences it's hard to dismiss all of them as delusional people seem to be able to feel their Consciousness as part of something greater or part of can communicate with somebody else so I I think that I wouldn't take a hard and fast view on Consciousness as it's always just me because I would want to you know problematize the word me what exactly am I I am part of I am part of first of all um the society uh that formed me the the culture that formed me the culture to which I hope I contribute I'm also part of nature nature is in me it's not around me like the environment it's in me and I'm in it and I would say because we don't have to go there at all but but I believe that actually very importantly there is um a Divine or sacred aspect to the cosmos and that in that we also exist so I I would you know take take a different view as if were from the kind of atomistic view of the eye yes I mean there's um again it was it was merely a a proposal I don't quite know where I land on it because I'm I'm also quite taken by the work in this case of of SRA and um more laterally Kenneth willerford on the fact that what we call subjectivity may just be conscious and just knowing s so um Kenneth wilford's got this really nice paper I think 2016 in one of met journals about how you know Consciousness is this bizarre phenomenon where there is this phenomenal feeling that it's for a subject but every every time we've tried to find that subject whether it's neurophysiologically or in the humean introspection we can't find it and so what does it mean to be for a subject that itself might not be present and he says that well that the subject relatum of Consciousness might just be the Consciousness episode itself yes and I think that's um a potentially quite neat solution yes to this kind of vexing issue but more broadly about Consciousness is the river and I really like I I too enjoy this metaphor because it also brings to mind this this relationship between change and Oneness and and brings to mind herac ex his insights makes me think firstly about it's consciousness's relation to matter because I is it would you say well matter is by nature somewhat atomistic um self- distinguishing and cut off and it seems to me that you know the whirlpool has uh it although it's part of nature it's part of the water it also has this kind of self-d distinguishing way of being and so is there a way in which Consciousness interacts with matter such that in me it's taking this form in some intricate manner that I I can't quite understand because I don't know the relationship between matter and Consciousness and I'm

sure no one does but in some ways matter is fixing Consciousness at least for a a transient period of time such that it can manifest in a in a thing like me hello all uh before letting Dr Miguel Christ answer my question I wanted to very quickly jump on and let everyone know that when I was formulating this question what I had in mind was actually a talk that uh Ian had given about three years ago on the relationship between m own Consciousness and the reason why I want to make this very clear is because it might seem that in formulating that question like it's my idea it's my theory and I want to express that it's not and it's totally my fault for not having referenced the talk and not having made it clear that actually this was Ian's uh original Theory and idea uh the reason why I formulated this question is because obviously a lot of people might have not seen the talk and I wanted to lure and get I to talk about this incredibly interesting idea uh but I think unfortunately the execution it may have seemed that I was trying to take credit for it and I'm really really not so I wanted to come on before the podcast goes live because the last thing I want to do is take credit for ideas that aren't mine plagiarize especially when it comes to someone like Ian who has worked so hard for so many years to establish these incredibly rich and insightful and thought-provoking theories so all of this to say that this notion of uh matter as the kind of stabilizer facility Ator um yeah of Consciousness is not my theory and so if it came across like that when I was asking this question I really wanted to just come on the record and say uh it's absolutely not it's Ian's what I will do is I will link this wonderful talk that he gave it's about an hour and 50 minutes it's on YouTube I will link it in the video description below but anyway the reason why as you'll see Ian basically recapitulates what I've said is because obviously these are things that he had already spoken about uh so yeah totally my fault but wanted to come on here and clear that up because again I don't want anyone shouting at me I don't want anyone getting the wrong idea and I definitely don't want to be taking credit for ideas that aren't mine so with that being said I hope you enjoy the rest of the conversation and I will now leave you with Ian because he is far more interesting and wise than thank you what I think about the problem of matter and Consciousness as we think of it is that frankly despite enormous efforts over very long periods of time nobody has been able to give a coherent and convincing explanation of how something like Consciousness can simply emerge from matter unless matter somehow already partakes of Consciousness or is in Consciousness or is an aspect of Consciousness if it has nothing to do with Consciousness then no amount of hand waving and gesturing towards complexity can really make something like Consciousness the inwardness of things come out of it um and In the comparison between how we know Consciousness and how we know matter

Consciousness is primary in that we can only know matter via Consciousness I may only know Consciousness via matter the brain but I don't know that it's certainly not necessary but it is necessary that to know matter um and by the way we've never seen matter nobody has ever experienced matter all we have experienced is aspects of our Consciousness that have qualities that we call material and I think that what what matter is is a phase of Consciousness much in the sense that water goes through phases it's flowing and and transparent and runs across your hand it's it's opaque and hard and can crack your head open it's it's in this room there's tons of UN believably in this room and if there weren't I wouldn't live and I can't see it so conscious people say well it doesn't look like matter well no it wouldn't do but um it doesn't have to what matter is doing I think is providing two things one of them is permanence for a while a degree of resistance which brings permanence so it's resisting change and that brings permanence and it's also resisting is containing and providing some sort of bound to what I can do and what I can think and without any bounds without any um the thing becomes chaos and without any definition for creativity to happen there must be an opposing force that is overcome and anything to be capable of lasting it has to have this degree of resistance to change and that is what matter is in my view wonderful yes there's so much here um that comes to mind I mean very briefly I guess a materialist neuroscientist here would might pulling their hair out and going well just because you can't know ever there's just an epistemological limit that means we can't ever know for sure that the brain creates Consciousness that needs to be held as distinct from an ontological premise that the brain causes Consciousness so it might be I never know this I can never prove it because of David Chalmers and Nagle and and all of the philosophical arguments that have been leveraged in this direction but nonetheless um on on logically speaking it's the best solution we have um that they seem so coupled and the brain is so complicated and it's doing all these wonderful things of course Consciousness just Springs out because the moment you shut one part of the brain off or you have a stroke or a lesion uh cons some element of Consciousness disappears do you find any truck in that uh distinction between the epistemology and the ontology here not really because um perhaps of VI will disappear if I change the settings on my screen um parts of experience will disappear if you if you go around injuring the brain no no problem there the but I don't show I don't believe I believe there are only three possibilities frankly that the brain is either an emitter of Consciousness which seems to be the the favored one amongst um reductionist materialists or it is a transmitter of Consciousness or it is a permitter of Consciousness in other words it somehow shapes and filters Consciousness and makes it my

Consciousness and that's what I believe the brain is and I know you weren't making an argument there but you gesture towards complexity but you know as you're perfectly aware the cerebellum has 80% of the neurons in the brain and is massively more complexly interconnected within itself with pinji cells being one of the most um massively ramified and interconnected cell neurons and and yet it doesn't sustain Consciousness on its own so I I think it's not like that I mean I had a conversation with um Mark SS the other day and and Mark says you know there's this yes place in the in the in the tectum I think maybe the tegmentum but I think it's the tectum of the midbrain where you know two cubic millimeters of white matter if you put a needle in and and disturb it then the person's gone but I don't think he thinks and he said he didn't that that little bit of matter creates the Consciousness it can't do well he's Mark's been on the podcast before and I like Mark's I like Mark's work a lot and I think it's very insightful I guess he's a he'll call himself a dual aspect monist um and so there's some underlying thing and Consciousness is one aspect of it and matter is another it seems to me that you're leaning slightly towards a more pansus position but it it's almost a different form of pansism from let's say a traditional um well I guess beron Russell and his monism it comes to mind here because Russell and laterally Philip gof who's the kind of premere um panus at the moment this idea that well as you say if I take um you know it's very akin to what you're saying about never having seen matter if I have this cup of water um it's not that the matter is the cup of water right its fundamental nature is consciousness and matter might just describe what um what physics does but not what physics is now is there a difference there between what you're saying which is the conscious matter is a facilitatory thing or process rather um that allows for creativity and the resistant forces that lead to self-actualization versus saying well no no what matter actually is is consciousness is are those two different things well I'm naturally a bit skeptical of grand pronouncements like that mat is consciousness but I think it's an I think it's a way in which Consciousness can manifest um and perhaps that's not a huge difference um I'm having a conversation with Philip go go as well so I'm interested in what he will he will have to say about my view but my feeling is that he and I um probably come quite close on this so yeah it is kind of have you much about sorry on yeah no no no I mean again to lay my class well I'm quite a part I'm I'm quite party to a pany but that might just be because I I don't know think yeah no I I look I I guess I don't know oh one thing I do push back against is the epistemic hubris of thinking that we know um and we certainly don't what have you thought much about Hydro enic children and what that might say for this relationship between matter and Consciousness because these are children who's uh such an

enormous portion of their brain is absent and yet they seem to be behaving in a conscious manner now again a materialist reductionist might say Well they're not conscious and um they might employ some kind of zombie argument what what what do you think of the kind of ramifications of that well I I would again you know reject epistemic hui here and say I'm not sure but but um we to make it clear for um any viewer the what we're talking about is not hydri in which there is quite a large part of the cranium filled with cerebrospinal fluid instead of with the brain tissue that would normally be there but I think you're referring to hydran and ke in which children are born with virtually no brain at all so things stop at the brain stem that's right um and what is quite extraordinary and I don't know how to explain it is that in the absence of any kind of um visual cortex they are able to recognize and apparently see and distinguish people in their faces they like pieces of music and have preferred pieces of music they have preferred toys um and I think once you've seen one of these children I mean of course nobody I mean you can be dayart and say I don't know that what I see out of my window are really people they might just be machines walking about with hats and coats I mean it's always open you to take that position but I want to appeal to a degree of intuitive imaginative common sense about it that it's very unlikely that that that is the case it doesn't feel like that and I wonderful photos of them these photos of them sort of at Disneyland smiling while their while their siblings are on their lap and um yes I think Common Sense might pril but again humility first and foremost there's a um the one thing I wanted to ask about when you're talking about resistance is the importance of negation in your work yes um so you I think I believe you quote Spinoza who says all determination is negation and I think you say that um there are two things one can do in this world is either say no or not say no which I think is fantastic um because the word yes just doesn't even intervene but what does that again going back to sort of Consciousness and our way of being how does that play out in terms of our experience what what is it is is this a metaphysical fact that in terms of things uh self-individuating there's is a as you say a fundamental tension that if if the you know you know if if became LAX would there would we would just dissolve into the soup and active inference for reasons like we can talk about agrees with this or or is this also about our conscious experience that in some ways that which means that it doesn't dissolve into the Super and just becomes about everything intends about everything is the fact that Consciousness in some ways is saying no and also permitting things to enter into its field yeah um I suppose what I was getting at was that first of all negation is how anything becomes defined as what it is after all defining is laying down boundaries and in a way something only

comes into being through the being bounds to it if it is boundlessly mixed with everything else there's only one thing it can be which is God or the Eternal being or something but anything that we know in this world has to be defined and this process is creative it's how you know when a when an artist is putting down paint on the canvas he or she is making a decision to paint this and not something else and it's a very simple point that you know when Michelangelo created his David he didn't put anything together he just said no to the stone all over the place until finally he'd got the thing that he wanted um and you could say he was saying yes to the the image I suppose but I suspect that from what one knows about the Act of Creation that the thing actually becomes what it is in the process and it is an evolving relationship between the artist and the artifact or the work of art so and and what I was suggesting is that when we see things seeking I give the image again of water because it's so useful for so many things but the image of the river you know apparently seek its way through the landscape but all that what is happening is that the landscape is either saying no or not no to the water it can't say yes to the water it's not in a position to do that it either fails to stop the water or it stops the water those are its choices um but that actually brings into being the river and I think that this is an image of how um almost anything comes into being that in the in the first instance it's through the absence of negation to something and let's see where this goes and what happens and one the thing is I suppose um achieves a certain degree and there isn't an absolute off Point um of actualization then it is more possible for us to respond to it as it is and wish to further actualize it yes again it brings to mind draus um because draus has this idea that sort s of when one is uh actualizing action you never have the the conditions of satisfaction in your mind you're just it's just unfolding and I well but interestingly draus does talk about and I believe this is based off Merlo Ponte that there's a kind of tension in that act and I think he uses the word tension and by that he means um there's a if one senses that one is moving away from an optimal grip on the situation one is always drawn back to to what you know to an expert way of being um and again I think this just it really does speak to the fact that there is no conscious affirmative yes in that it is just sculpting away and and and and again it comes in some ways back to Aristotle I mean Aristotle's idea that you can only judge a life once it's been lived it's it's a disclosure it's an unfolding it's not um it's not purposive in the sense of a kind of cartesian removal and this is what I want to do and I'm going to do it now and no this is this is this is not quite right and N said um the purpose of a Melody is not to reach its final note but unless it reaches its final note noted as a Melody so yeah I mean I'd love to go back if we can to self-consciousness because I think it's so important and it also

leads to schizophrenia in which there's a kind of hyperconsciousness that is the person's existence in a way but what you know I don't know why but when you were giving those instances I was thinking of something I described in the matter with things which is to do with expertise again. It is a topic that in the early stages of learning expertise it's useful to have things to guidelines to follow rules to observe but that when you get to the the last two of his five stages of expertise having such rules or principles is actually damaging it limits your ability to develop your skill and you remain forever mediocre and one of the things I'm intrigued by is people who have motor skills such as I I I I out of the blue the The Physician who looks after the um the TT races on the Isle of Man you you'll know what I'm talking about but um maybe for the International audience there's this completely crazy and infinitely dangerous sporting activity that takes place once a year on the Isle of Man off the coast of Britain during which all the islands roads are closed and motorbikes race around a 62 mile course that goes over ordinary public roads um round tight bends over Brakes and you know down sudden inclines and so forth and at certain times they they're speed can approach 200 miles an hour so this is both very dangerous given this is not a perfectly smooth track um and um remarkable but the thing that comes out of it is that they enter a sort of mental state in which they have a kind of total awareness but if anything distracts their attention to a detail then they they lose the grip and a lot of it is to do with not trying so you know and this is a familiar story isn't it that one Rider um felt he was doing very well and then he realized that something was happening to his bike and he had to stop and fix it and he stopped and fixed it and after that he just poled on because he thought there's no point in in going fast now because I know I've lost the thing actually he turned out to have the winning speed so well there there's a there's a paradox in there I think this is Slingerlands trying not to try um and I'm not quite sure how to resolve that Paradox yes but it's fascinating on I I hate to I hate to do self aggrandisement and self advertisement but actually very fitting we're talking about flow States because I just uh myself and several others including KL freston just posted this uh just published this paper on Flow States under an active inference framework um and self-awareness in flow States uh which was just published a couple of days ago in Frontiers so you may you may find that to be um that up but perhaps perhaps slightly frustrating we try we try and provide a kind of Bayesian model for FL states and and perhaps draus would be screaming at me because he'd be saying oh you can't mathematize it um but really interesting something that's very interesting about dra's skill hierarchy he has this example about airport um about fighter jet pilots and that when fighter jet Pilots instruct novice fighter jet Pilots there's a

checklist of uh you know an order so you must check the light and but when they actually I believe it was Ice scan experiments on fighter jet pilots in the Jets they realized they were not following that order whatsoever so it's not just that the rules are um embedded in coded unconsciously followed out it's almost that they're not followed out which implies that it's um they in expertise perhaps and maybe this is not the case for all expertise but draus is quite strong that chess play expert chess players are not following rules um there's almost a relinquishing of rules and there's there's actually quite a strict division between those who have to kind of follow this epistemic Behavior this guideline following and then those who just do in some sense and again I don't I don't think D has comes to the conclusion of exactly how that works but it's very interesting to think that this isn't just a uh absorption Act of of absorbing rules but actually might be an absolving of rules it might be a den of R I'm not quite sure what that means very it's fascinating none interesting I mean I hope that what goes on a lot of the time is the absorbing of procedures so that you're not even aware you're doing them but you're you're suggesting and and have no reason to doubt it that at times they're not followed um I mean it comes back to the observation that you have to really really know the rules before you're allowed to break them and I think that is very important because um too much of the time we neglect skill in in in um allowing people to they want to develop a talent and what we do is encourage them just do what you feel like and let it all hang out and you know I know where that comes from and some of that is good but actually something very important is also much more important which is you really do need to learn the procedures you know by the time Raphael was nine he could master all kinds of techniques that most people would be drawing for decades before they'd be able to do so but he did actually learn them he didn't just wasn't born with them so there's that um but yes I mean when when that pilot landed the plane on the Hudson and when a surgeon suddenly reacts to whatever is going on in theater I mean when asked afterwards they often say well I don't know what I did I just did what I have to and and it is interesting that that may actually involve denying certain rules I mean obviously there are certain rules you know why they're there and in certain and in most contexts they're important but in a certain particular context it might be a crucially waste time wasting step to follow that rule or procedure and you need to cut through to something else yes well it speaks to our our bureaucratic infrastructures which I'm sure we we can speak about at length I mean one thing I will will add is that we we don't at least in this paper we don't deny the the presence of anything we we don't just act as if um you know well there would be no model if it was just this kind of elusive mysterious unfolding this disclosure I

think what the really important thing that we we tried to emphasize was context and the importance of contextual cues and so we argue that a lot of what is happening in flow States for example is this constantly unfolding contextual queuing which we sort of can understand as a habitual mental action and draf talks about this a lot as well that even if it's not okay uh given this contextual cue I do this it puts you in this kind of gestal condition where you know exactly what to do without knowing that you know and it's all implicit and it's unfolding and yet if exactly what you said about the distraction if a distraction is sufficiently uh takes you away if it doesn't act as a contextual cue for this unfold these unfolding Dynamics then you're out of the state and you have to work your way back into it by in some ways recovering quite a loaded philosophical term these contextual cues so I think all of this is very much um on on on track and I think we're finding convergence here I guess another part of this is intuition yes and the role of intuition in in in skillful coping and expertise what do you what do you take intuition to be and how much do you think that it drive lives our everyday Behavior because as drers and Mer Ponte and Hy would tell us we live intuitively right I know how to sit on this chair I know how to speak English I know how to drink water is there um how how can we integrate intuition into this picture yes well again I'd sort of reverse that and say how do we integrate the rest into the picture of intuition which is the the overarching picture I mean if you think about intuition is basically the unconscious processes that we have we can now put a fairly accurate figure on this that all that our nervous system and our body in general is aware of um is infinitely greater than the amount that we self-consciously are aware of so the little bit we're self-consciously aware of that you know that you know is the bit that you're looking at with the left hemisphere uh focal attention but that is um estimated to be slightly over half a percent out of the 100% the rest is all being done uh outside the glare of attention and you know sorry this is a bit kind of um simplistic I guess but I hope you'll forgive me um I think people say the unconscious in a lot of people's minds it's something like the cellar it it's somewhere down there as it were in the dark that is you know a tank below everything that's going on in the sunlight and occasionally fish pop up from the bottom tank and occasionally boat go down but instead I would use the image of a a spotlight on a stage so where the spotlight is is what you're focused on but it hasn't altered the rest of the stage all the rest of the possibility is still there it's just not actually currently in the spotlight and I think that's a better way to think of it now what intuition is really saying is that what it means is to contact that other 95% in whatever way and it's not inferior I mean of course it can be mistaken but as I um repeatedly say um simply following a line of argument can lead to mistakes as well as in

the schizophrenic for example so you need context and you need intuition as well as you need the power of logic and not dismissing any element in there so what I think intuition enables us to do is the complex stuff so it enables us to make decisions with far-reaching consequences to balance alternatives to um to to to form an intention to do a certain thing we don't do these things consciously they come up through periods when we are not attending and what they have over the explicit Consciousness is that in that explicit Consciousness if I articulate what I'm thinking we can only have a sentence or serial sentences but in intuition there may be 12 or 15 strands that are contributing and when I try to articulate I automatically go back to the obvious I I articulate what seems to be Salient but all the things that are not Salient but nonetheless very important are speaking to me with the result that I can't remember his name but there is an academic jurist at one of the max plank institutes in Germany um who says he's the director of it and he says people who are in charge of large organizations should be telling their people to use their intuitions and I think he's dead right I mean there's an army of um psychologists no disrespect to the profession who are going around making a lot of money telling people in organizations not to trust their intuitions but I think this is very very much a half story and the worst half of the story because by not listening to your intuitions you end up in very strange places I think there are examples around us of how that is happening so in the matter with things I have a whole section of three chapters on intuition and the literature on intuition and how it works and whether it works and so on and once again um it's not an All or Nothing thing but the when we overthink things we usually make them worse unless unless things have already broken down in other words it's like a toolkit that you get out of the boot when you need to repair and this is where Whitehead said this wonderful thing that operations of thought are like Cavalry Charges in battle they can only be made at decisive moments they require Fresh Horses and they they be the main thing you rely on so this is this is how it is we shouldn't be thinking that by overthinking we will make better decisions we often [Music] don't well I I think to add to that um I complete I'm in complete agreement I think the other I slightly ironic thing is this claim that you could ever dispense with intuition um and and I think if you try to dispense like this is I think in many ways it comes to what you were saying about the the take that the non-conscious is a kind of uh um discarded Sellar that very rarely sees the light and I think in some ways we can blame Plato for this and some Two Worlds mythologizing but my my my take on this is that for to use your Spotlight example and as you know it's a very common example in the attentional literature it's often taken that this this Spotlight is is s it's moving itself it's like the the hand is within the spotlight my claim is that

no no no this Spotlight is fun Mally enslaved again to use this word by so much that you can never become aware of so even the very fact of paying attention to a siren or it is not an arbitrary act to pay attention to a siren even if it means even if it's purely reflectional why is it reflective well it's probably because at some point there was some evolutionary pressure to pay attention to things that are loud or luminous and whatnot and I think there's been a great mistake in dichotomizing these things exogenous endogenous unconscious conscious such that we don't realize that they're always in play with one another they're always in concert and if you think that your Spotlight is solely uh guiding itself you are very blind to how biased that Spotlight is and how much it is driven by that which looks underneath your conscious awareness so I think I think again it's it's it points exactly to what you said in all of your books which is relationality that it's there is no one the other and in many ways a self-guiding spotlight would be totally useless to what end apart from the spotlight own maintenance would you know it has to be in some ways in constrained and subserving and and being a very good Emissary but I worry that the spotlight attention metaphor is itself emblematic of this kind of self-referentiality that the left hemisphere is yes is propagated and you know one of the things I ought to just mention is that people worry that if um things happen through unconscious promptings that somehow they are not deciding but this is because they've already decided that they are just that little tiny bit of conscious them as though all their unconsciousness was not also a very important part the the main part of who they are and it it's you know that's the right right yes well again it's the great irony that even that part of you which is conscious is not blind or uh encapsulated with respect to the parts that are non-conscious they're constantly filtering information in and out of one another so even if you wanted to say oh I'm just this little moral that's well you can't be because it's kind of like heraclitis is River it's always in flux um and excellent okay this is all super I want to very briefly ask about this the relationship that we've been speaking about with respect to the hemispheric differences and and and the underlying cognitive infrastructure to broader social changes historical changes um so when I was you know when I finished the master in his emary I I like reading book reviews after reading books because I I'm always interested to see whether my thoughts converge with with other peoples and I thought there was an extremely um slightly shallow one by Owen Flanagan um so this is 15 years ago so you may not remember he he he even gets some the he gets it yes he gets it mixed up but he makes see he makes the claim that um 2,500 years of History are ill understood as our hemispheric differences and then as a result trying to do the mapping between hemispheric differences and historical evolution is a slightly fra problem I'm

curious about just generally and it could be brief what do you what relationship do you establish there between the brain and historical events whether it's the Greeks the Romans The Romantics and so on I can't help reflecting that um Flanigan's University Duke just invited me back to give the John P McGovern lecture I looked for him in the audience he was hiding no um there can be a misunderstanding because I mean very literal-minded people go but a brain is something in a human being it can't be in a society and you know okay I get that but what I'm pointing to is not either that changes in the brain have caused changes I I'm not in the business of saying what causes what and I don't think the brain itself is causative of such changes in society but what I what I thought thought about it was really inspired by Lou Sassy's work particularly Madness and modernism and you know this extraordinarily brilliant book I think in which he looks at the patterns of thinking and creativity and art in the last 100 years or so and uh relates it to the phenomena experienced by people with schizophrenia and obviously it's not that we've all got schizophrenia it's that as he hints in one of his appendixes um schizophrenia mimics what it's like when you have right hemisphere dysfunction and so what he's really saying is at the moment our society ignores all the kinds of more subtle things that the right hemisphere knows in favor of fatally simplified theoretical understandings of things and that this is causing problems so I thought of thought well H so but presumably this well clearly this hasn't always been the case so when looking back in history what do we see and and you know in a in a former life I was effectively interested in the history of ideas and um so I I thought well let's review Western Civilization from the Greeks through the Romans up to The Rance and and Beyond to the Enlightenment and Romanticism and modernism and what I think I can see there in three cases is that there's been a a change in emphasis um over time uh in which in each case there is initially a rather sudden and surprising um period it tends to happen rather quickly in which suddenly there's a move forward of both the right and left hemisphere together working well complementing one another always with the left hemisphere being the as it were the functionary that assists the master rather than taking charge that's when things go astray but as as each civilization has advanced in terms of its its wealth its power its extent this has um gone hand in hand with an extension of the left hemisphere's um control because the left hemisphere's resold DET is in fact control Power utility um and as result the subtler things that had sustained the civilization began to wne and the civilization crashed so what I'm talking about there is is not that I mean obviously the brain is always Evol evolving so there will be changes if you were to brain scan a Kean man in the 6th Century BC and and and somebody now but they'll be very minor and they're not what I'm

talking about what I'm talking about is how we use our brains our brains are there still and are right hemisphere is telling us things but we've been somehow indoctrinated and I'm not suggesting that there's some you know cult that's indoctrinating is I'm just meaning the mindset that we've got in through through an industrialized Society and a capitalist Society is that everything is to do with the massing power and material it's it's all about utility and that everything else is sort of going to hell in a handcut effectively um but it's not that we lack the Insight so what a lot of people who read my work either the master or the matter um say to me um is you you made me aware of things that I knew at some level but had no way of articulating or or or you know arguing for but but you've given me that yes I mean that I mean I think that's the great import of your work is again it comes back to this idea of um of revealing that which was already there so I I never when I so I haven't read the matter of things uh and I will but when I read the master and his Emissary I didn't I mean you you you're obviously introducing a hemispheric theory I don't I never felt that you were imposing anything I never felt it was an imposition on my way of understanding it was more of a um as you say this kind of um this unraveling this un- you know the way the liit talks about knowledge is it's just chipping away to to leave something that was already there I mean I think an interesting um Revolution that's going on moment is both mindfulness and psychedelics um and to be uh oddly personal I'm in Mexico currently and Mexico is uh particularly replete with Sil cybin and um on a Sil cybin trip a couple of weeks ago quite an intense one and this may in part be because of the my reading of your work it was this remarkable multiplicity and unity unity multiplicity but I wonder whether I wonder what whether what you thought about this which is to what degree will the people who are reading your books who then go out and and see uh you know the universe in a in a petal to paraphrase William Blake um to what degree do you worry that it's this is just a sort of biasing effect of having read your work and it's not actually a kind of authentic disclosure of the world's structure at least on their part but rather a sort of again a kind of left hemisphere Eric self-evidencing self-confirming uh way of being because that was in part my worry over the experience it was incredibly profound and beautiful and I was in the middle of the forest and things were felt very much alive and and and had their own autonomy and yet were partaking in this kind of broader structure but after it finished I thought well maybe that's because I've been reading a lot of Ian's work um and so is there a way of potentially mitigating this this uh this potential well I have nothing to say as you know about um mind altering substances of this kind um it's interesting though that you think there was some cousinhood between

whatever it was you were reading and this experience I like that and I don't know what I think about that I mean people often say is aren't these substances sort of releasing the the right Hemisphere and I don't think they are I think what they're doing is they suppressing frontal cortical uh function and the frontal cortex of course is the major filter of what it is that we do attend to and experience and I think that that's when that's removed then there is as the flood gates are open and and you see a lot of things together and maybe you get a sense that yes they belong together which is I think probably a real feeling that they do we can't go on leading our lives in that state whereas I hope that we can go leading our lives with um whatever it is I've given um to people because they do often say things like you've altered my life once you see this you you can't not see it and and they say for the better you know I they report moving things about their way their lives have changed through reading my work which which is completely unforeseen I had the slightest idea I hope it would be interesting to some people I had Great Hopes but had no idea that it might actually have this effect on people which is just totally lovely um obviously I I I don't think that what I'm suggesting is is is um is false or farfetched I think if it were probably it wouldn't have the impact that it generally seems to have even on people who are fairly skeptical to to begin with you know because most people in in particularly in the world of psychology I think are there a sort of Catch 22 that when you're told that hemisphere differences are not important um because they don't really mean anything then of course you don't do the reading to find out whether they are or not so it's a self-confirming situation and one of the things I'm battling against is um the the tendency to fail to report um hemisphere differences I'm hoping that in future when people do brain research they will always even if it's a null finding you it just report are there any hemisphere differences it's a question that really must be asked and the more we ask it the more we will understand what I've tried to understand no doubt inadequately but you know we will understand it better well you're your I mean the master his Emy and I'm sure the matter of things is obviously replete with references nonetheless so people can go there but I I I agree I mean I think in many ways as a kind of budding psychologist and philosopher it's it's as you said it it's a consequence of the Frameworks in which we adopt and so if we have a particularly popular framework and let's say something like um top down versus bottom up differences and everything is studied or or or or pursued through the lens of hierarchical systems and as you say for example caraman and tki very interested in in in this verticality and you know to the to the detriment of horizontality and well I I I think hope y you'll you you'll start to see the fruits of your labor um because I think I do think the conversation is around

hemispheric differences is is is changing quite immensely talking about Gran unifying theories I asked people uh on Twitter very briefly a couple of days ago about what they wanted me to ask you okay and um you know this is as I said it's it's an active inference Insight podcast although I spend so much of my time talking about Consciousness it ought to be a Consciousness podcast but people wanted to know about what you thought about this kind of grand ifying Theory and whether you'd thought about it at all so have you given Carl's work much reading uh have you spoken to Carl and have you got any IDE you know any thoughts on the free energy principle or or predictive process processing or anything you know aligned well I feel that I'm slightly like molter's cook in in this you know molier when he was trying out his plays he used to read them to his cook first to find out what the reaction was um and I'm afraid I'm as ignorant as B cook and I have I have tried to understand and I I hope I have understood some of what the active INF principle involves and the free energy principle I I'd like to be told really what what does it add to something that we have always believed which is that no perception is naive that we don't we're not just like sort of photographic plates but we go to the world with um predictions with presumptions about what it is we're likely to see and so on and that's of course very beneficial um it not only saves energy but it saves time and it makes us much more effective at um living but what what does this really add to that because I mean yes we we change our behavior in the light of that p and the behavior makes the environment feedback I mean really what it what it seems to me to be saying in a rather more complicated way is what I'm saying is that everything is relational that um what we what we perceive and take to be reality some of it comes from us some of it comes from out there these that doesn't make it unreal and that the environment if I must call it that but the world around us and ourselves are interdependent we are bringing one another about so there's a constant dialogue between them that's the way I see it and I find that the the the mass and all that is very impressive but I'm not sure quite what it is um perhaps you can tell me what it is it's adding I want to say so Carl what is the question to which the answer is the free energy principle okay so if I if I can speak on his behalf which I can't but I shall I I think I think I actually think it'll be a wonderful conversation with you with you and car and others but the idea here is that well there's an ongoing debate is what I'll start with there's an ongoing debate between the instrumentalist and realist depictions of active inference so on one hand you say okay we can model the behavior of everything through this basian framework so everything that persists through time as the thing that it is can be described in terms of basian inference and what this means is is that things embody in code a model of themselves and the world and

their relationship and they act or they can be described as acting in order to provide evidence for that model and this can be taken from um this as I said the SC I think this is the one of the important things of this is a totally scale-free principle so you can leverage this to describe the existence of and the persisting existence of anything from an atom to the cosmos now that's the instrument of this thing which is it's an extremely useful modeling tool because it talks about how things can be sparsely coupled to one another and thus how they change in relationship to one another exactly what you've been talking about such that if I act on the world and the world acts on me we enter into this mutually coupled econ Niche agent relationship which again is No not not different from the insights of ecological psychology or um embodied for cognitive science now the other the the the the the complete other end of the spectrum is the realist version of Act of inference which in many ways takes a predictive processing account which is that the brain cannot is not just that it can be described as encoding a generative model it is a generative model and the way all the brain cognition is or all forms of brain cognition are fundamentally predictive excuse me such that we leverage predictions we leverage predictions about the world and when the world when we receive the data from the world which is ultimately always Divergent from our predictions we either update our model of the world or we act to change the world so as to um make that sense data more concordant with our generative model and so in principle this is a updating of Helmholtzian um percept his notion of uh perception as inference But it includes action and I think that's the great part of it which is that um we in many ways Enslaved the World by changing it in accordance with our preferences now obviously the world is also enslaving us right because if a car hits me doesn't matter what I want from the world that car has provided evidence for itself as a kind of bulldozing mechanism but I think what a lot of people are interested um from you is again like the maths is the maths and people are trying to understand exactly how um this can be implemented computationally I guess the question for you is what is the role of a kind of meta Theory or or or were the limitations of a meta Theory and as you say is this just a reordering of something which was implicit in all these other strands of theoretical biology Neuroscience physics and so on or is it actually saying something new so I really think ultimately this is just telling me that you should speak to Carl um well I'm sure it would go but I'm curious about you very Molly cook here um I I it I've given you Theon Pub version fine I mean that's maybe maybe what I need um I mean I I still feel that I'm not aware of what it is that it's added to my my understanding of how perception and action go on uh it hasn't changed the way I think about it maybe because I haven't understood it or maybe because I've

already intuitively understood it and Incorporated it I don't know but um the thing that it does do though for me is to bring with it a feeling that comes from the sort of overwhelming idea of this meta Theory or meta world or meta whatever that somehow somehow we've mastered this we can predict it it's all mechanical it it produced an image of a world that I don't recognize and I don't really want to know whereas I'm more Blake like I go out and I see these wondrous living beings and and all that and the beauty of the world and the complexity and the unpredictability of it seems to have been lost in a sea of graphs and sort of you know calculus I I I not that there's anything wrong with it I would want to say but I think it needs to recognize its own partiality that it's just a partial yes approach that's what I'd say about the mattera thing and I don't claim by the way that you know well no I just going to say no no please I don't claim that the hemisphere theory is a sort of meta Theory though I believe it has many many consequences if it's true and the evidence for those consequences is something that I'm you know looking for both in the um scientific literature and in the social psychological literature yeah well I guess one thing to add to this is the active inference or the free energy principle cannot well some people might think it can but it can't solve Consciousness and so um we're still left this massive but I guess I guess to to to finish up to return to the hemispheric differences you finished the the the master in his Emissary with this with this very like provocative also quite uh alluring and endearing line which is that um if it all turns out to be a metaphor then that's fine because you have high regard met which I liked a lot and metaphors really um underpin for me the they underpinned the the tenor the the the frust of that whole book and and and again holding this tension that Mets allow us to understand the world but also um in many ways if if it is just a metaphorical Theory we still need to pay respect to that and also return back to the phenomena yes what um just in terms of the hemispheric differences what I I was slightly confused by what it actually meant for that to be a metaphor what would it mean for the hemispheric theory to be a metaphor well what it what it wouldn't necessarily mean is that um there's anything wrong with the science um Science is based on metaphors um the important point is that you can understand anything without a metaphor and um that that the um the use of metaphor in science and philosophy is is particularly um important and heavy so you cannot step outside metaphors if you wish to understand something it's very obvious in in physics and it's been argued for by various philosophers um so I think that's what I mean I meant there it was a slightly Sly one in which I was kind of I was giving a little obeisance at the end and saying well maybe I got it wrong but nonetheless actually you can only understand through metaphors and I've given you a pretty important

metaphor but but rather literal-minded people have gone oh well oh so he thinks it's only a metaphor like only a metaphor meaning there isn't any literal truth to it but there is literal truth to it I mean I'll I'll you know stake a lot on that and I've taken I hope you will well you know what you do with your time is up to you and it's a terrific ask to read this book that matter with things but I think after this I'm certain you will enjoy it but in it I take the the discussion to Greater levels of dabs and you think I've put a lot of evidence into the master in his emistry and the bibliography there looks like a a a postscript um the bibliography here is is 180 Pages because I really needed to look at a lot of stuff and I really did look them up and inspect them because a lot of the time it's amazing working in in in in the area is that you look up something that's been quoted and requoted and quoted as saying X and you look them up and actually it says not X it doesn't even say why not so there are one or two which that has happened but uh yeah so I do care I care about empirical evidence I think it's incredibly important and if we lose that what have we got you know to I don't say empirical evidence is everything of course it has to be taken up into an understanding and so forth but if we actually throw away the importance of empirical evidence we're cast free we have no loadstone to to guide us and one of the thing Sinister things that that's happening now you wanted to talk about what's happening now very briefly but I mean one of the things that worries me very badly is the well-known fact that science is now being heavily skewed by political pressures to conform to a certain narrative about the way things are and that um perfectly good science that suggest that it's more complicated than that is not being recognized and when Nature actually writes an editorial saying it's not going to publish things just because they're good science they have to be on message well that's the end of science as far as I'm concerned do you with that in mind do you hold you know people talk about the pendulum the the political pendulum swinging and I guess in this case it's the the scientific pendulum swinging do you think nature and I mean nature both you know both in its literal sense of Nature and and nature is in the journal will will nature bite back at nature you know will will will this will will this come home to R because I think there is this uh utopian I guess there's this utopian idea that if we can be if we can self-censor if we can confirm the theories that we already have in mind without questioning the underlying axioms of those theories then everything will just be Dandy do you think um that reality itself always comes home to roost and we bite back well they do if we give it long enough to do so without destroying ourselves I mean in the longer run reality will come back of course and utopianism has been the cause of the most cruel despotic totalitarian projects in human history so no we must resist that at all costs and and I suppose what is

beginning to happen is that people are starting up universities private universities where people can really learn and study without being forced into a straight jacket about what they say and journals will spring up that will publish things and so on and no doubt Twitter will ban them or something but we've got to get Beyond this kind of mind that we we ban things if we don't like them the answer is not that the answer is yes to be able to look at what is said in a reasonable rational temperate Manner and discuss it because if we could do that there is nothing that cannot be resolved discussion is better than fighting I profoundly believe well indeed I think um again this is I think a nan or Doan idea that you can't get rid of things by pushing them underground in many ways you just make them more vicious um yes and curious inverse way I saw that with religion in the Soviet Union um I've some I mean although I I suppose I'd have to call myself of a religious cast of mind I've always had a kind of LoveHate relationship with the church but when I was in Russia in the Soviet years in the 70s I remember going to the uh cathedrals in what was then Leningrad and Petersburg and seeing how they thrived you know middle of Wednesday morning on a weekday the the cathedral was packed with people praying lighting candles singing together and so on I thought this is this is so bloody moving because also as they came out there were these leather jacketed sunglass thugs photographing them all so that they could make sure that their children would never go to university so it took courage you know and when you repress it it thrives now that is no longer repressed I don't know I've heard that congregations are falling off maybe wrong well that there a return to a I don't like the word spiritual but a um a spiritually minded way of being right I mean in many ways one has to give credit here to to Someone Like Jordan Peterson um and I know you've spoken to to John vvi I think John and you and John Share quite similar perspectives here with a skepticism on organized religion but also recognizing the loss that a fundamental reality and being 25 years old and having grown up in Metropolitan London my whole life um I can tell you that the urge in me is to be extremely cynical and condescending towards those of a religious mentality because it's always been the way I was raised you know but I think I think there's a there's a counter and whether this and I guess the interesting point is the metaphysical n grounding of this is some what well it's not arbitrary but it's it's it's it's not as important as the fact that church or um these centers are points of communion I know and I I think we dispense with that our our Peril I mean I found for example that actually team sport is the most unbelievable um form of healing in some ways just to just to be in concert with people to be trusting people to in flow with people and um yes I I think I think this actually all points to your work that uh as you say relations

precede relate that we are we are kind of nothing without relationships this is well this has just been an absolute Joy I am so very grateful to you I'm very grateful for the Mexican WiFi for having only messed up once um I'm on the border of Mexico I'm on the border of Mexico in Biz and I have a boat to take to visit all places so the Wi-Fi there I think will be particularly spotty so I'm um I'm very grateful for time um I always like to ask people to just say where people can find them uh find their work I mean obviously you have these two enormous and epic and wonderful toes but beyond that where can people find what you're working on currently I mean what are you working on currently well uh you won't find that anywhere actually because I'm not really sure um I've got a number of number of things that are bubbling away um I may not have time to finish any of them but um basically the place to go is channel mcris if you just um put channel mcris into the Google bar it'll take you there um some people think it's an imperative but it's just a description and uh there's uh YouTube you can always just Google my name on YouTube and I'm afraid there's an appalling number of interviews on there my friends complain that they can't go anywhere i' well I'd like to I think no you're one of the great podcast guests thank you I don't I I I genuinely do mean this um some of the podcast no well some of them you've done are just wonderful um I particularly liked your first one with Kurt J mongle yes where you're sort of guiding him his slightly uh yeah he he it was incredibly intense now he's he's read the matter with things we're planning to do another so that's good but he wanted to make sure he he'd read it properly well good well he's he's also got some overlap with the active inference Community actually because he's interviewed Carl a number of times and um I think he he hosted some round tables so so the worlds do Collide and converge but um I'm going to read the matter of things um I'm going to make that I'm going to make that a project okay very good um let's see how long it takes no hurry okay thank thanks very much thank you hly thank you enjoy being in Mexico it's a lot warmer than it did here on Sky I imagine but good luck yeah I've heard it's rather chilly on the British AIS so I'm I'm quite happy despite their electrical issues all right byebye wonderful all right so much bye thank you